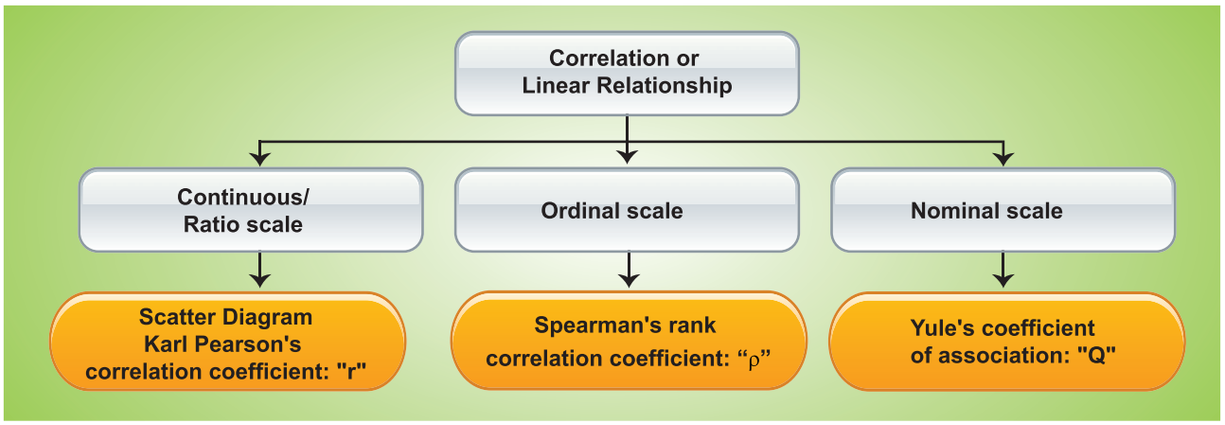
Statistical tool selection



Methods to find correlation

1. Scatter diagram
2. Karl Pearson's product moment correlation coefficient : '*r*'
3. Spearman's Rank correlation coefficient: 'ρ'
4. Yule's coefficient of Association: '*Q*'

NOTE

For higher order dimension of nominal or categorical variables in a contingency table, use chi-square test for independence of attributes. (Refer Chapter 2)

4.3 SCATTER DIAGRAM

A scatter diagram is the simplest way of the diagrammatic representation of bivariate data. One variable is represented along the *X*-axis and the other variable is represented along the *Y*-axis. The pair of points are plotted on the two dimensional graph. The diagram of points so obtained is known as scatter diagram. The direction of flow of points shows the type of correlation that exists between the two given variables.

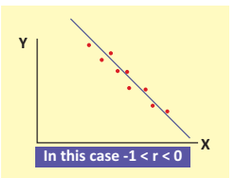
1) Positive correlation

If the plotted points in the plane form a band and they show the rising trend from the lower left hand corner to the upper right hand corner, the two variables are positively correlated.



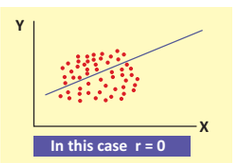
2) Negative correlation

If the plotted points in the plane form a band and they show the falling trend from the upper left hand corner to the lower right hand corner, the two variables are negatively correlated.



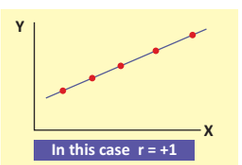
3) Uncorrelated

If the plotted points spread over in the plane then the two variables are uncorrelated.



4) Perfect positive correlation

If all the plotted points lie on a straight line from lower left hand corner to the upper right hand corner then the two variables have perfect positive correlation.



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